FOOLING YOUR BRAIN

The algorithms that guide our thought processes are often flawed and open to exploitation

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If you've been reading the cover stories in sequence, by now you should've got some understanding of the way the human mind works. In fact, you'd even know that the mind, the brain, the subconscious, the soul (?), consciousness, and awareness are all different things. They are interlinked of course and they all contribute in equal measure toward making you... you!

Call it the hubris of being the dominant species on the planet, but we often think of ourselves as perfectly rational beings. We think our decisions are taken purely based on data or valid inputs, and processed through a seemingly dispassionate processor. Even if we aren't dispassionate about a topic we tell ourselves that we are aware of our inherent biases and therefore unaffected by them. The truth is though, our decision making heuristic models have been shaped over long periods of time through evolutionary processes and are meant to work properly as a shorthand in very peculiar or specific scenarios. Often these scenarios involved scarcity and matters of life and death. This "lizard brain" of ours has trouble when dealing with abundance of resources, large data sets, and the kind of one-step removed social interaction we have today in the digital realm.

In the language of computing, these are all attack vectors by which these vulnerabilities of our internal coding can be exploited. Here's a look at some of them.

1. ILLUSION OF CHOICE

No one likes to be told what to like. Well, unless it's Apple doing the dictating (we kid, we kid).

Those in the business of selling things have found that offering a couple of obviously wrong "options" pushes consumers to the option that they wanted to sell in the first place. Take subscription services offered by AV companies, cloud storage or even streaming services like Netflix. You'll find that the middle plan is often always the best value for money. This isn't a coincidence. The plans are structured in a way that the consumer feels they are getting the better of the service provider by making the right choice. We evolved to take as much advantage of a situation as possible. Reciprocal altruism only works when there are likely to be repeat encounters in the parties so when you see there are three choices, your lizard brain tells you to opt for the seemingly best one quickly. Almost urging you to jump at it.

> A fun exercise to understand the concept of reciprocal altruism and the evolution of trust: ncase.me/trust/

Choice can be "weaponised" in other ways too. Research suggests people are more likely to do something when they are offered a choice. For instance a boss might tell you, "either do the data entry or file the field reports, the choice is upto you." This would work far better than just telling you to "do the data entry".

2. FOOLED BY "EVIDENCE"

Patterns are very interesting. We are programmed to find them. During the early evolutionary stages of our species, our survival depended on making sense of patterns, understanding them and making predictions based on them – it was a matter of life and death.

Patterns that have a causal relationship in their occurrence are very useful but often there is no such relationship. When we observe a set of data points and link them together to derive meaning we call it coincidence. Carl Jung went a bit further and coined a specific term for it – synchronicity. You know how if you read about a particular concept – say half life (not the game, the nuclear physics term) – you see it popping up all over. Is the universe conspiring to show it to you more? Or the frequency of the term is still the same you are just more likely to "catch" it? It's most certainly the latter. The observer-expectancy effect and confirmation bias have been known to ruin many scientific experiments because our mental coding is a little too biased to catching patterns. In an experiment conducted by Whitman and Woodward, the researchers found that presenting evidence one after the other thereby creating a pat-